FORMATTING INSTRUCTIONS FOR ICLR 2025 CONFERENCE SUBMISSIONS

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ABSTRACT

1 METHOD

Our system extends a pretrained zero-shot TTS model with control over non-verbal vocalizations (NVV) in Vietnamese. Since Vietnamese speech with NVV annotations is scarce, we synthesize training data by inserting voice-converted NVV clips into real Vietnamese recordings at natural pauses, with an inline tag at the matching position in the transcript (§1.1). We then fine-tune the TTS model on this data mixed . At inference, an inline tag such as `[laughter]` in the input text produces the corresponding vocalization in the cloned voice. Figure 1 gives an overview.

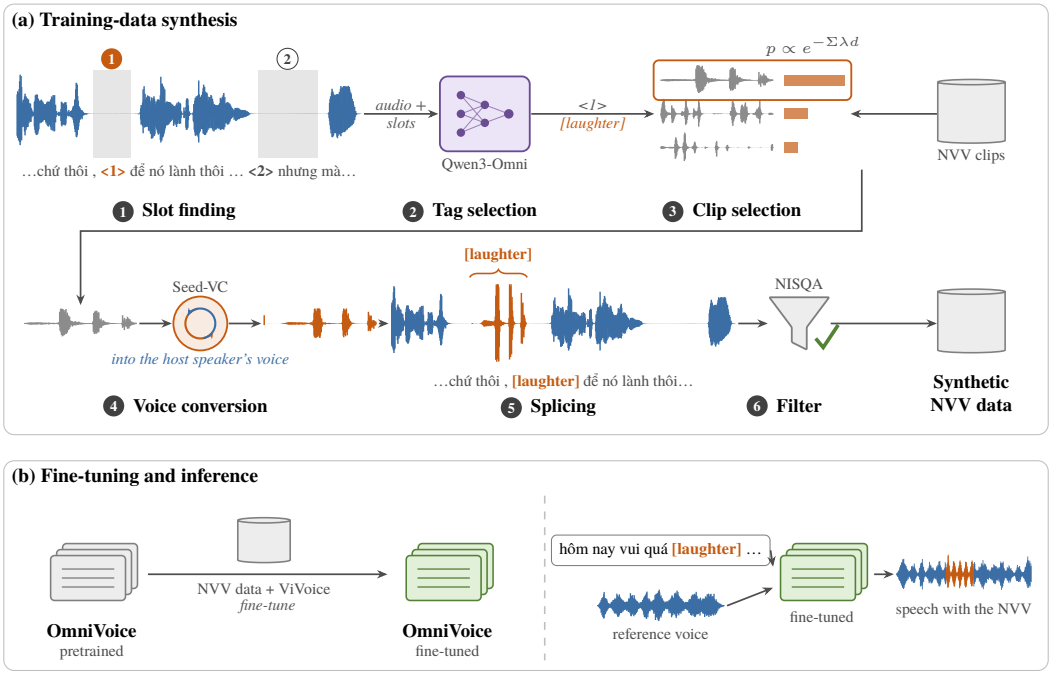


Figure 1: Overview. (a) Training-data synthesis on a real example: VAD pauses between aligned words become slots; an omni LLM picks one slot and one tag; an NVV clip of that class is sampled to match the speaker, converted to the speaker's voice with Seed-VC, spliced into the pause and quality-filtered. (b) OmniVoice is fine-tuned on the synthetic data with untagged ViVoice; at inference an inline tag produces the NVV in the cloned voice (waveforms in (b) are illustrative).

Table 1: NVV clip pool per tag. *Temporary*: counted on the local build of the clip index, which holds the full VocalSound corpus and a partial NVS-38K run; regenerate from the server's index before submission. Hours are the summed clip durations. Sources: VS = VocalSound, DNV = Deeply Nonverbal Vocalization, FSD = FSD50K, NVS = NVS-38K.

Tag	Sources (clips)	Clips	Hours
`laughter`	VS 3,498; DNV 25	3,523	4.16
`sigh`	VS 3,498; DNV 51	3,549	4.48
`cough`	VS 3,497; DNV 50	3,547	4.49
`throatclearing`	VS 3,497; DNV 54	3,551	3.78
`sneeze`	VS 3,496; DNV 52	3,548	3.78
`sniff`	VS 3,499	3,499	3.88
`breathing`	FSD 324; NVS 55	379	0.11
`screaming`	FSD 233; DNV 51	284	0.14
`gasp`	FSD 147; NVS 17	164	0.03
`snoring`	FSD 161; NVS 2	163	0.08
`crying`	FSD 83; DNV 24; NVS 11	118	0.07
`yawn`	DNV 48; FSD 22; NVS 2	72	0.05
`moan`	DNV 45; FSD 23	68	0.05
`pant`	DNV 44; FSD 11	55	0.04
Total		22,520	25.12

1.1 NVV TRAINING DATA

1.1.1 NON-VERBAL VOCALIZATION CORPUS

VocalSound (Gong et al., 2022) alone is limited: it covers six classes (laughter, sigh, cough, throat clearing, sneeze, sniff), and its crowdsourced clips often carry long silences or stray speech from a loose cut. To add classes and keep only clean events, we pool NVV clips from three kinds of source: (1) other NVV corpora: the Deeply Nonverbal Vocalization dataset, whose vocal classes we use as they are, and FSD50K (Fonseca et al., 2022), whose clip-level labels we localise in time; (2) NVV events cut from an existing tagged speech corpus, NonVerbalSpeech-38K (NVS-38K); and (3) events cut from speech generated with our transcripts by commercial TTS systems (FishAudio, Gemini TTS). For VocalSound we keep only the test and validation splits (§2.1).

Each source locates its events differently. For NVS-38K, whose captions carry one inline tag, we force-align the caption words with Qwen3-ForcedAligner. The aligner tends to stretch the neighbouring words over the non-verbal sound, so the gap between the two words around the tag does not give the event's extent. It only confirms that the tag refers to the dataset's own detected non-verbal region (within 1 s), and that region is what we cut. FSD50K has clip-level labels only, so FlexSED's frame posterior for the class prompt locates the events (frames ≥ 0.5, gaps up to 0.3 s merged). Clips that carry two of our target labels are skipped, since their events could not be told apart.

Every located event is then cut and checked in the same way. Its edges are tightened to the frame energy within ± 0.2 s of the region, then padded by 75 ms and faded by 10 ms. The event is kept only if (i) FlexSED scores its class ≥ 0.5 over the sound and scores no VocalSound class higher, so a new class does not duplicate an existing one; (ii) no other labelled sound of the recording scores ≥ 0.5 over it; (iii) Qwen3-ASR finds no lexical word in it, so no speech overlaps it; (iv) its length is within a per-class range (e.g. 0.2–2.5 s for breathing, 0.8–4.0 s for a yawn); (v) its SNR against the recording's noise floor is at least 6 dB; and (vi) at most 0.1% of its samples are clipped. Table 1 lists the resulting 14 classes.

1.1.2 SLOT FINDING AND TAGGING

Following SynParaSpeech (Bai et al., 2025), we first find the pauses where an NVV could go and only then let a model choose among them. Placing the tag first and aligning it afterwards fails on connected speech: the boundary between two words is usually continuous voicing, and on a 20-utterance pilot the nearest real pause was a median 2.02 s away from the aligned tag. Silero VAD finds the speech regions. Regions less than 0.10 s apart are merged, so stop closures and glottal

breaks do not count as pauses, and the remaining gaps are trimmed by 0.05 s at each edge and kept if at least 0.12 s long. Silence before the first word and after the last is ignored. ViVoice transcripts have no word timing, so Qwen3-ForcedAligner aligns the untagged transcript to the audio. A pause becomes a *slot* when it overlaps the gap between two aligned words; if two pauses fall in one gap, the longer is kept. Each slot records its midpoint and the character offset of the word after it. If any word cannot be matched back to the transcript in order, the utterance gets no slots and is skipped, rather than risk a tag between the wrong two words. The slots are marked with numbers in the transcript, e.g. "chiáº¿n lÆ°á»£c , <1> tá»©c lÃ <2> nhÆ° váºy".

The recording and the marked transcript go to Qwen3-Omni-30B-A3B-Instruct, served locally with vLLM. The model is asked to listen to the delivery and answer in JSON with the one marker where a non-verbal sound would most plausibly occur and the one tag from the offered list that fits what the speaker is doing there, or `{"slot": null}` if no position suits any sound. We sample with temperature 0.2 and top-p 0.95. An invalid reply is retried, up to three attempts in all, and a null answer skips the utterance. The model never writes transcript text: the tagged transcript is built in code by inserting `[tag]` at the chosen slot's character offset, and the splice time is the slot's midpoint, so text and audio refer to the same point by construction.

Left alone, the model settles on a few neutral classes; on the pilot it chose only laughter and sigh. A tag balancer therefore keeps a running count of the tags assigned so far, carried across runs, and withholds from the offered list any tag that holds more than 25% of them, with the share taken over at least 20 tags. A reply that names a withheld tag is re-asked. The tag is offered again once the other tags catch up.

1.1.3 Clip selection

Within the chosen class, one clip is drawn at random, weighted towards clips that match the host speaker on the properties that Seed-VC passes through unchanged. Speaker timbre, pitch level and pitch contour are rebuilt by the conversion (with F0 conditioning off), so we do not match on them. Clips whose peak is below 0.05 or that are shorter than 0.35 s after trimming silence, and clips with more than 1% of samples at full scale, are excluded. Four measures are taken on each trimmed clip and on the 2.5 s of speech before the cut: *intensity*, the mean of the clip's crest-factor and duration ranks within its class, against the rank of the context window's RMS among all windows of that length in the utterance; *clarity*, the SNR as the ratio of the 90th to the 10th percentile of 20 ms frame RMS (0–60 dB, measured on the whole utterance for the speaker); *tempo*, envelope peaks per second, compared in octaves and used for laughter only, since the timing of a cough or sneeze has nothing to do with speaking rate; and *tilt*, the slope of a line fitted to the log envelope, in dB from start to end. Each distance d_a is divided by a tolerance set to about a third of its measured spread (0.35 rank, 12 dB, 1 octave, 18 dB), and a clip is drawn with probability proportional to

$$w = \exp\Big(-\sum_a \lambda_a d_a\Big), \tag{1}$$

with weights $\lambda = 1.0, 1.5, 1.0, 0.5$ for intensity, clarity, tempo and tilt. Clarity weighs most because VocalSound's microphones vary far more than its speakers' loudness. Tilt has half weight because its premise, that a speaker trailing off should be followed by a decaying event, has not been confirmed by listening. Sampling, rather than taking the closest clip, keeps every usable clip possible, so a soft-spoken speaker still sometimes gets a loud laugh. For the same reason the intensity target is drawn per row from $\mathcal{N}(r, 0.2^2)$, clipped to $[0, 1]$, around the speaker's own rank r. Clip features are precomputed once for the whole pool, so every row samples from the full class.

1.1.4 Voice conversion and insertion

The selected clip is trimmed of silence, peak-normalised to 0.95 and zero-padded to at least 1 s, as Seed-VC's input chunking requires. It is converted with Seed-VC v1 (50 diffusion steps, classifier-free guidance rate 0.7, length adjustment 1.0, F0 conditioning off), using the host utterance itself as the voice reference, at 22.05 kHz, and is trimmed again afterwards. F0 conditioning is off because its pitch tracker (RMVPE) is built for speech and singing and fails on coughs, sniffs and throat clearing.

The converted event's RMS is set to the RMS of the 2.5 s of speech before the cut plus a per-class offset (laughter $+3$ dB, sneeze $+4$, cough $+2$, throat clearing -2, sigh -4, sniff -5, and 0 for the

other classes), and its peak is capped at 1.5 times the speech's 99.9th-percentile level. The recording is cut at the slot's midpoint, and a 50 ms exponential fade (curvature k=5) is applied to all four edges: the end of the speech before the cut, both ends of the event and the start of the speech after it. An exponential fade reaches half level within the first 14% of its length, which keeps the sharp onset of a laugh or cough that a cosine fade would soften. Each side of the event should then carry about as much silence as the speaker normally pauses for. The target pauses are

$$\begin{aligned} T_{\text{pre}} &= \bar{T}_p \cdot s_g \cdot s_{\text{pre}}^{(c)} \\ T_{\text{post}} &= \bar{T}_p \cdot s_g \cdot s_{\text{post}}^{(c)} \end{aligned} \tag{2}$$

where $\bar{T}_p$ is the median gap between voiced stretches of the utterance, clamped to $[0.04, 0.40]$ s, s_g=1 is a global scale, and $(s_{\text{pre}}^{(c)}, s_{\text{post}}^{(c)})$ are per-class scales. Events that end on emptied lungs need a longer beat after them: laughter $(0.6, 1.4)$, sigh $(0.5, 1.2)$, cough and throat clearing $(0.7, 1.0)$, sneeze $(0.8, 1.3)$, sniff $(0.5, 0.7)$, and $(1, 1)$ for the other classes. The target is a total, not an addition: the inserted silence is $\max(0, T - T_{\text{exist}})$, where T_{exist} is the silence already on that side of the cut, so the speaker's own pause is never shortened. Inserted silence is filled with the recording's own background noise, its quietest 100 ms tiled with crossfaded seams, since generated white noise audibly hissed. Finally the whole recording is peak-normalised to 0.95 and written as 16-bit PCM at 22.05 kHz.

1.1.5 Quality filtering

Every finished row is scored with NISQA v2.0 (Mittag et al., 2021), a no-reference predictor of the mean opinion score (MOS) and of four quality dimensions, on both the source recording and the spliced one. A row is kept if (1) its NISQA MOS drops by ≤ 0.5 from source to result, (2) its NISQA discontinuity drops by ≤ 0.22, catching an audible seam at the splice, and (3) its boundary activity at the cut is ≤ 1.2. The thresholds apply to the drop rather than to absolute scores, because an absolute floor mostly measures how clean the input corpus was. Boundary activity covers what NISQA cannot hear once the fades and the silence have smoothed the junctions. It is measured on the source recording as the RMS of the 50 ms on each side of the cut divided by the RMS of that whole side (0 means the cut is in silence, about 1 means it is in speech of average loudness), and a row is judged by its worse side. The thresholds were set on the 20-utterance pilot: the MOS threshold removes one clear outlier, and the other two were chosen by listening. Seed-VC's diffusion is not seeded, so a rejection means that one conversion was bad, and re-rendering a rejected row may produce a passing take.

1.2 Model adaptation

We build on OmniVoice (Zhu et al., 2026), a massively multilingual zero-shot TTS model that maps text directly to multi-codebook acoustic tokens with a non-autoregressive diffusion language model and clones a voice from a reference recording. The released model cannot generate Vietnamese NVVs as it stands, for three reasons. First, its tag set is small: of its 13 inline tags, only `[laughter]` and `[sigh]` are non-verbal sounds, the other 11 being lexical interjections such as `[question-ah]` and `[surprise-wa]`, so 12 of our 14 classes (Table 1) have no tag at all. Second, it has never seen a tag in Vietnamese. About 8.5k hours of Vietnamese speech, ViVoice among it, are in its training data, but all of it is untagged; its tagged data come from Chinese and English corpora (NVSpeech, NVS-38K and ParaSpeechCaps). Third, it was not trained for tag control: these corpora are a small fraction of its 581k hours of training data, and the OmniVoice paper does not evaluate tag control. In practice the released model often drops the tag. Given Vietnamese text, it produces no vocalization at the tag for XX% of `[laughter]` and `[sigh]` tags, the two it does support, and for XX% of the other 12 (NVMR, §2.4 and Table 3).

We therefore fine-tune OmniVoice on the synthetic data of §1.1, which supplies all 14 tags on Vietnamese speech, mixed with untagged ViVoice utterances to preserve ordinary Vietnamese synthesis (§2.1). Nothing else changes: the training objective is OmniVoice's own, and at inference the tag is written inline in the input text, as with OmniVoice's tags.

Table 2: Synthetic NVV training set per tag, after quality filtering. Each utterance carries one event. Hours are the total duration of the finished utterances.

Tag	Utterances	Hours
laughter	XX	XX
sigh	XX	XX
cough	XX	XX
throatclearing	XX	XX
sneeze	XX	XX
sniff	XX	XX
breathing	XX	XX
screaming	XX	XX
gasp	XX	XX
snoring	XX	XX
crying	XX	XX
yawn	XX	XX
moan	XX	XX
pant	XX	XX
Total	XX	XX

2 Experimental setup

2.1 Data

Training uses ViVoice capleaf (2024) and the synthetic NVV set of §1.1: XX utterances (XX h), each with one event from the 14 classes of Table 1; Table 2 breaks it down by tag.

VocalSound clips are drawn from its test and validation splits only (5,446 clips, about 907 per class, with no speaker in common), so a vocal-sound classifier trained on VocalSound's training split can still be evaluated fairly on our output. The other NVV sources are used in full. Each training batch mixes synthetic and untagged utterances at a ratio of XX:XX. We hold out XX speakers, unseen in training, for evaluation. All audio is resampled to XX kHz.

2.2 Training

We initialise from the public OmniVoice checkpoint (k2-fsa/OmniVoice) and fine-tune all parameters with AdamW (β_1=0.9, β_2=0.98), peak learning rate XX, XX warm-up steps and decay, for XX steps with a batch of XX s of audio per GPU, in bf16. Training takes XX h on NVIDIA B200. The training objective is unchanged from OmniVoice. We select the final checkpoint.

2.3 Baselines

We compare against (i) OmniVoice without fine-tuning, given the same tagged input; (ii) OmniVoice fine-tuned on ViVoice alone, without NVV data, which isolates the effect of the NVV data from in-domain Vietnamese adaptation;

2.4 Evaluation

Test set. The test set has 420 real Vietnamese sentences, 30 per class, each with one tag. Each sentence is chosen so that the tag fits its content, e.g. "bật cười" (burst out laughing) for laughter, "hoảng hốt" (startled) for gasp, "ngủ say" (sound asleep) for snoring, and "hạt tiêu" (pepper) or "trời rét" (bitter cold) for sneeze. The tag sits where the sound would naturally occur, after the action word or at a comma. About 65% of tags are mid-sentence; throat clearing and breathing lean towards the start of the sentence, and snoring and sniffing towards the end. A few sentences were lightly edited to fix spelling or cut a garbled ending. Sentences are 8–30 words long (median 16), with no digits and almost no English words, so the WER below is not inflated by the ASR writing out numbers or mishearing foreign words. Each sentence is synthesised once for each of XX held-out speakers, prompted with a XX s reference recording and its transcript, giving XX items per system.

Synthesis. Every system uses OmniVoice's voice-cloning inference with the language set to Vietnamese, and the same item gets the same random seed in every system. The released inference code tokenises an inline tag apart from the text around it, whereas training tokenises the whole string, so a model can miss or garble tags it has learned. We therefore tokenise the whole string at inference for every system, the released model included, so that all systems receive the same input.

Objective metrics. NVPA, WER and SS follow Track A of the VLSP 2026 NVV TTS task. Each metric is reported on its own, as the mean over all items of a system unless stated otherwise; no combined score is computed. Higher is better for every metric except NVMR and WER.

NV placement accuracy (NVPA) checks whether a vocalization of the tagged class occurs where the tag sits in the text. The tag is removed and the remaining words are force-aligned to the 16 kHz output with a Vietnamese wav2vec2 CTC model (`nguyenvulebinh/wav2vec2-base-vietnamese-250h`) and torchaudio's `forced_align`. Each word boundary is a slot, and the tag sits at one of them. Qwen3-ForcedAligner is not used here: besides stretching words over non-verbal sounds (§1.1), on Vietnamese it collapses runs of words to one instant. FlexSED then scores each class prompt over the whole recording, in 10 s windows with a 5 s hop, and each frame keeps its highest posterior across windows. A class's events are runs of frames above 0.2 that reach the class threshold at least once. The threshold is 0.5 for most classes, 0.4 for throat clearing, and 0.7 for breathing, which FlexSED also scores at 0.5–0.7 on pauses where no breath is heard. Runs less than 0.12 s apart are merged, events shorter than 0.08 s are dropped, and events shorter than 0.25 s that touch either end of the clip are ignored as cut-off tails. A tag is *correct* if an event of its class lies within 3 s of its slot with no aligned word in between; a word that overlaps the event by at most 0.05 s does not count as in between. Otherwise the tag is *misplaced* if exactly one word separates the nearest event of its class from the slot, *wrong class* if only another class fires at the slot, and *missed* if no event of any class is at the slot. Classes are scored strictly: a sigh, pant or gasp also triggers FlexSED's breathing prompt, but counts as correct only when its own class fires. NVPA is the share of correct tags (0–1).

NV miss rate (NVMR) is the share of missed tags (0–1): the model made no sound at the tag, either skipping it or reading it out as words. NVPA counts a missing, a misplaced and a wrong-class event alike as failures; NVMR isolates the first.

Word error rate (WER) is computed with a Vietnamese Zipformer (`sherpa-onnx-zipformer-vi-2025-04-20`) using greedy search on the 16 kHz output. The reference and the transcript are normalised alike: tags are removed, the text is put in Unicode NFC and lower-cased, and every character other than a letter or digit becomes a space. WER is computed at corpus level over all items of a system, not as a mean of per-item WERs, and reported in percent.

Speaker similarity (SS) is the cosine between the ECAPA-TDNN embeddings (`speechbrain/spkrec-ecapa-voxceleb`) of the output and of the speaker's reference recording, both at 16 kHz and L2-normalised (0–1 in practice).

Subjective metric. The only subjective metric is MOS. XX native Vietnamese listeners each hear one sample at a time and rate its overall quality on a five-point scale (1 = bad, 5 = excellent). Each system contributes XX samples, and we report the mean MOS with a 95% confidence interval.

3 RESULTS

Table 3 compares the released OmniVoice with the two fine-tuned systems. For context, its upper block lists the results that [CITE] report for Vietnamese TTS systems on their own test set.

Table 3: Released and fine-tuned OmniVoice on Vietnamese NVV synthesis. *Shared*: `[laughter]` and `[sigh]`, the two classes the released model has a tag for; *New*: the other 12 classes of Table 1. Every system gets the same tagged input, so for the new classes the released model sees tags it was never trained on. MOS, WER (%) and SS are over all test sentences; MOS is given with its 95% confidence interval.

	NVMR ↓		NVPA ↑				
System	Shared	New	Shared	New	MOS ↑	WER ↓	SS ↑
viXTTS	–	–	–	–	2.51	3.27	0.7021
F5-TTS	–	–	–	–	3.93	3.02	0.8196
VietVoice-TTS	–	–	–	–	3.58	2.37	0.7834
VieNeu-TTS	–	–	–	–	3.69	2.00	0.6524
OmniVoice (released)					XX±xx	XX	XX
+ ViVoice					XX±xx	XX	XX
+ ViVoice + NVV (ours)					XX±xx	XX	XX

REFERENCES

Bingsong Bai, Qihang Lu, et al. SynParaSpeech: Automated synthesis of paralinguistic datasets for speech generation and understanding. *arXiv preprint arXiv:2509.14946*, 2025.

capleaf. viVoice: Enabling Vietnamese multi-speaker speech synthesis. `https://huggingface.co/datasets/capleaf/viVoice`, 2024.

Eduardo Fonseca, Xavier Favory, Jordi Pons, Frederic Font, and Xavier Serra. FSD50K: An open dataset of human-labeled sound events. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 30:829–852, 2022.

Yuan Gong, Jin Yu, and James Glass. VocalSound: A dataset for improving human vocal sounds recognition. In *Proc. ICASSP*, 2022.

Gabriel Mittag, Babak Naderi, Assmaa Chehadi, and Sebastian Möller. NISQA: A deep CNN-self-attention model for multidimensional speech quality prediction with crowdsourced datasets. In *Proc. Interspeech*, 2021.

Han Zhu, Lingxuan Ye, Wei Kang, Zengwei Yao, Liyong Guo, Fangjun Kuang, Zhifeng Han, Weiji Zhuang, Long Lin, and Daniel Povey. OmniVoice: Towards omnilingual zero-shot text-to-speech with diffusion language models. *arXiv preprint arXiv:2604.00688*, 2026.

A APPENDIX

You may include other additional sections here.